

HOANG BAO DUY LE

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EDUCATION

The University of Toledo, College of Engineering

Graduated Date: Dec 2022

Bachelor of Science in Computer Science and Engineering, Cumulative GPA: 3.67

Relevant Coursework: Artificial Intelligent, Computer Architecture & Organization, Senior Design, Computer Graphics, Systems Programming, Software Engineering

The University of Toledo, College of Engineering

Anticipating Graduation Date: Dec 2027

Master of Science in Computer Science and Engineering

TECHNICAL SKILLS & COURSEWORKS

Programming Languages: Proficient in Java, C#, C++, Assembly Language (80x86 & ARM), JavaScript, Python

Web Development: HTML5/CSS, JavaScript, PHP, React, NodeJS, MongoDB, Angular, MSSQL, ASP.NET

Network: Experience with networking and communications protocols (TCP/IP)

Others: Capable with Git, Blockchain, LLM, Machine Learning, AWS

PROFESSIONAL EXPERIENCE

System Software Developer | Nysus Solutions

January 2023 – Present

Coordinate with 20+ members (PMs & Developers) to create and improve custom web-based manufacturing software

- Maintain and update manufacturing software systems related to: real-time manufacturing execution system (MES), data collection systems, and web-based administrative applications, mainly involving integration to PLCs.
- Work with the Lead Application Architect to design and deploy a robust architecture that will facilitate growth of new applications being developed and deployed.
- Design and integrated MES software with MRP systems and floor equipment (e.g., PLCs, HMIs) to fulfill specific manufacturing requirements, facilitating automated production and streamlining information flow.

System Software Developer Co-op | Nysus Solutions

January 2021 – Dec 2022

Coordinate with 20+ members (PMs & Developers) to create and improve custom web-based manufacturing software

- Supported existing applications using tools developed and MS SQL Server Management Studio.
- Worked with Programmable Logic Controllers (PLCs) to automate and optimize industrial processes, ensuring seamless integration with existing systems for customers/projects such as DMS, DNUS, IAC.
- Utilized bar-coding languages to develop and maintain systems for bar code printers and scanners, enhancing data accuracy and operational efficiency for customers/projects DMS, Magna TEAMS/Norplas, Piston, ZF.

PROJECTS

Enterprise Automated AI Support

Sep 2025 – Present

Full Stack Developer

- Architected a coordinated framework with 7 specialized agents (Planning, Scanner, Frontier, Ensemble, MCP, and Specialist agents) that work in parallel to optimize response accuracy and speed.
- Implemented dual vectorstore approach using both OpenAI embeddings and open-source Ollama embeddings, enabling similarity search across 50,000+ historical support tickets with color-coded relevance scoring (90%+ similarity threshold).
- Built Model Context Protocol (MCP) agent that translates user questions into safe SQL queries, featuring automatic schema inspection, multi-database routing, and query result synthesis across 20+ production databases.
- Designed memory system with semantic search for frequently asked questions, reducing API costs by 40% and improving response time by 65%.

Ecommerce Website

Sep 2024 – Present

Full Stack Developer

- Develop a full-stack ecommerce platform with React and Next.js for a seamless and optimized user experience.
- Implement server-side rendering and static site generation with Next.js to improve performance.
- Build a scalable backend using Node.js and Express, with a MongoDB database for product and user management.
- Implement JWT-based authentication and role-based access control for secure user interactions.
- Design and integrate RESTful APIs for user authentication, order processing, and payment handling.